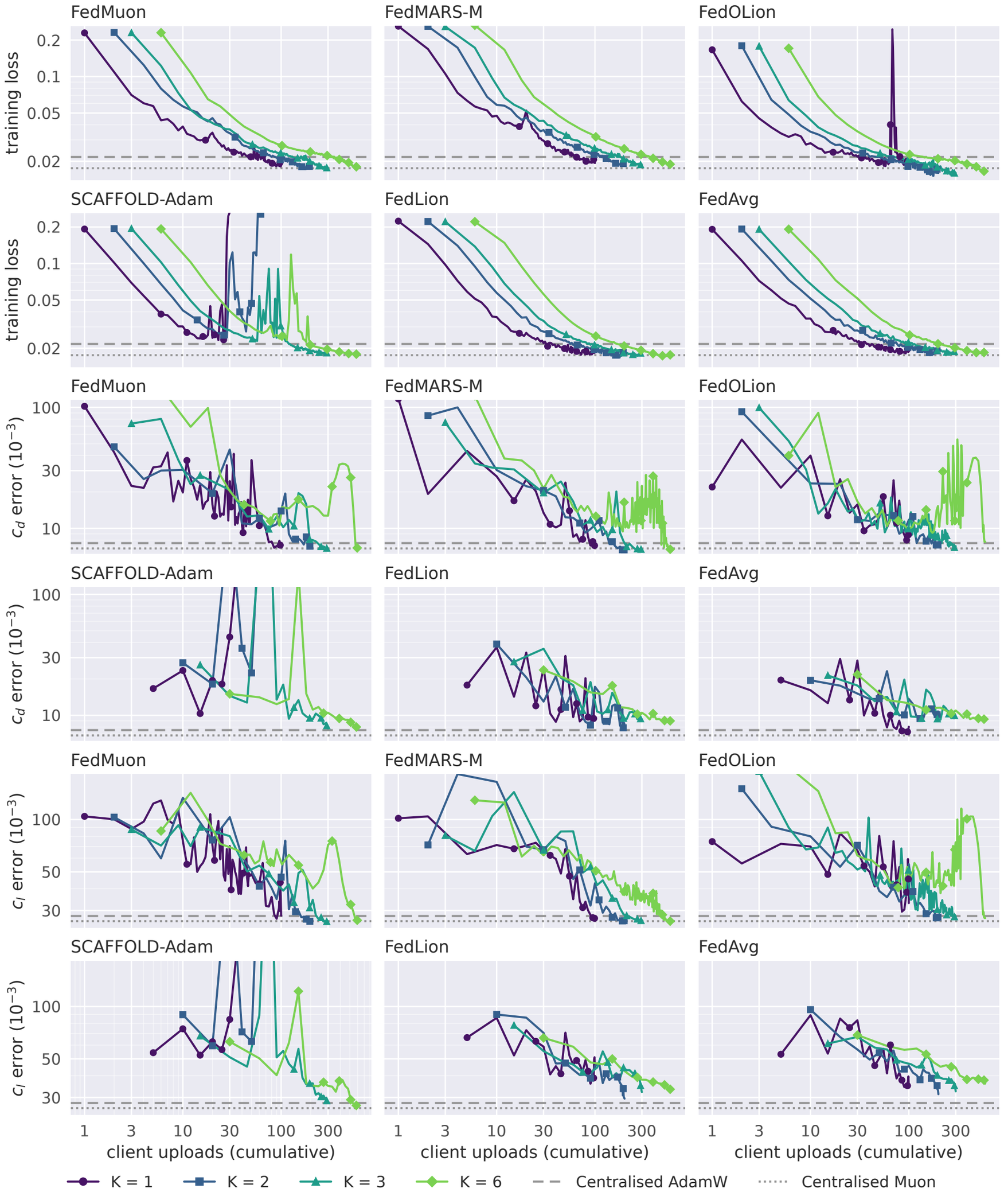

FedMuon
FedMARS-M
FedOLion
SCAFFOLD-Adam
FedLion
FedAvg
training loss
0.2
0.1
0.05
0.02
c_d error (10^{-3})
100
30
10
c_l error (10^{-3})
100
50
30
1
3
10
30
100
300
client uploads (cumulative)
K = 1
K = 2
K = 3
K = 6
Centralised AdamW
Centralised Muon